

Maxwell Plus: Turning to AI and Machine Learning to diagnose prostate cancer

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ABOUT MAXWELL PLUS



maxwellplus⁺

Using machine learning through Google Cloud, its own artificial intelligence-based Maxwell Plus processes more than 1GB of patient — across thousands of patients — to diagnose medical conditions such as

About Maxwell Plus

Founded in 2016, Maxwell Plus is a growing Brisbane-based medical technology startup that aims to rethink the way diseases are detected and diagnosed. The business combines machine learning with medical data including imaging and blood results to make the diagnosis of cancer and neurological conditions faster, more affordable, and more accurate. In late 2018, the business secured \$1.1 million through the Australian Government's Cooperative Research Centre Projects grant scheme to lead a collaboration to commercialize new AI

and Alzheimer's c
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algorithms on the platform to help detect and
diagnose Alzheimer's disease and other
neurodegenerative conditions.

Industries: Healthcare

Location: Australia

Google Cloud Results

- Enables the business to deliver results to clinicians in 1 to 5 minutes
- Scales to process 500,000 patient cases by the end of 2019
- Expands into AI-powered diagnosis of neurological conditions such as Alzheimer's disease
- Helps improve the

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Dataflow (<https://cloud.google.com/dataflow/>),

Google Cloud (<https://cloud.google.com/>)

(<https://cloud.google.com/spanner/>),

Cloud Storage (<https://cloud.google.com/storage/>)

(<https://cloud.google.com/kubernetes-engine/>)

Cloud AI & Machine Learning

(<https://cloud.google.com/products/ai/>)

Dataflow (<https://cloud.google.com/dataflow/>)

Cloud Spanner

Cloud Spanner (<https://cloud.google.com/spanner/>)

Cloud VPN

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

(<https://cloud.google.com/healthcare-api/>)

Cloud DNS (<https://cloud.google.com/dns/>)

Cloud Datastore

Cloud Datastore

(<https://cloud.google.com/datastore/>)

Cloud Load Balancing

(<https://cloud.google.com/load-balancing/>)

Cloud VPN

(<https://cloud.google.com/hybrid-connectivity/>)

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Cloud Healthcare API

(<https://cloud.google.com/healthcare-api/>)

TensorFlow (<https://www.tensorflow.org/>)

Google Workspace

(<https://workspace.google.com/>)

speed and
accuracy of
cancer
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allowing
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Effectively diagnosing and treating cancer continues to be one of the world's great healthcare challenges. According to Brisbane-headquartered medical startup Maxwell Plus (<https://maxwellplus.com/>), 3,500 people every day are diagnosed with prostate cancer alone — and 841 men die from the condition. Maxwell Plus founder Dr. Elliot Smith aims to change the way cancer and other diseases are diagnosed by combining machine learning and artificial intelligence (AI) with medical imaging.

Dr. Smith — who has a Ph.D. in Biomedical Engineering (MRI Systems Design) started Maxwell Plus to exploit what he describes as "a big gap in the market around automating the diagnosis of health conditions."

"Existing processes were typically manual and involved a lot of people," he says. "They do a good

job, but we have more people in need than we can care for comprehensively."

Dr. Smith believes machine learning and AI can remove obstacles to scaling out healthcare and improve the efficiency and accuracy of diagnosing medical conditions. "Our grand vision is to build an AI supported doctor anyone can connect to in order to obtain results affordably," explains Dr. Smith.

Maxwell Plus enables clinicians to submit blood data, patient demographics, and MRI scans to a machine learning-powered platform with AI capabilities to help diagnose cancer and other conditions. The platform uses machine learning models to deliver diagnoses faster, more accurately, and more affordably than existing systems. The business offers the diagnosis service to clinicians and platform for patients to engage and manage their risk. Men purchase an annual care plan or have one provided by their employer.

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*—Dr. Elliot Smith, Founder,
Maxwell Plus*

Maxwell Plus had successfully processed 50,000 patient cases by the end of 2018 and expected this to rise to 500,000 patient cases by the end of 2019.

Expanding to Alzheimer's

Maxwell Plus focused initially on improving prostate cancer diagnoses. However, in late 2018, the business secured \$1.1 million through the Australian Government's Cooperative Research Centre Projects grant scheme to lead a collaboration to commercialize new AI on its platform to help detect and diagnose Alzheimer's disease and other neurodegenerative conditions. Maxwell Plus is working on the project with medical imaging provider I-MED, Australia's national science agency

CSIRO, and specialist tertiary and quaternary (fourth in order) health service Austin Health.

According to Dr. Smith, the project will utilize a number of Google managed AI capabilities and test the Cloud Healthcare API

(<https://cloud.google.com/healthcare-api/>), which bridges the gap between clinical systems and Google Cloud (<https://cloud.google.com/>). Cloud Healthcare API supports the industry-wide DICOM standard for storing and exchanging medical images across radiology, cardiology, ophthalmology, and dermatology. This support for standards-driven data formats and protocols for healthcare technologies streamlines data processing through Google Cloud Platform (<https://cloud.google.com/>) services.

Starting on a single server

Dr. Smith and his team started Maxwell Plus — then known as Maxwell MRI — by running a number of functions and processes on a single server with graphics processing units (GPUs) and sizeable hard disk capacity.

However, this infrastructure could not scale to support the planned growth of the business. "Each case we process encompasses about 200MB of data in MRI scans alone," says Dr. Smith. "Once we include supplementary data, blood test results, pathology results, and genetic information, this load can reach more than 1GB of data per patient."

Maxwell Plus needed an infrastructure service that could scale storage and compute to support tens to hundreds of thousands of patient cases — without requiring heavy capital investment. "We wanted to move from accessing three GPUs to at least 30 GPUs without having to buy more servers or other associated equipment, so the cloud was the natural next step," says Dr. Smith.

Maxwell Plus evaluated public cloud services and determined Google Cloud Platform would be the best fit for its business. Managed services available through Google Cloud Platform would minimize its administration requirements, while Cloud Machine Learning Engine (<https://cloud.google.com/ml-engine/>) would help the business access as many GPU-equipped machines as possible to run its TensorFlow (<https://www.tensorflow.org/>) based machine learning training models.

Maxwell Plus started with some small experiments to confirm Google Cloud Platform met all its requirements, and migrated to the platform in February 2017. "We really started to scale up the data we had and consequently our computing requirements at that time," says Dr. Smith.

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Clinicians can now upload imaging and associated data to the Maxwell Plus platform running in Google Cloud Platform. This action triggers a range of upload pipelines that clean and standardize the data, before writing imaging data to Cloud Storage (<https://cloud.google.com/storage/>) and sending structured data to Cloud Datastore (<https://cloud.google.com/datastore/>) and Cloud Spanner (<https://cloud.google.com/spanner/>). The Maxwell Plus platform then converts the data into records that can be used to train new machine learning models or run evaluations through existing machine learning pipelines. The outcomes are then presented back to the clinicians and their patients.

The business is now running all its data storage and machine learning models in Google Kubernetes Engine (<https://cloud.google.com/kubernetes-engine/>).

"Google Kubernetes Engine allows us to scale fluidly in line with demand and the amount of medical

analysis we do," says Dr. Smith. "We have everything from our data storage to our machine learning models in Google Kubernetes Engine, which has been quite powerful for us.

"While our system incorporates a lot of traditional data access and modification, we also undertake a lot of asynchronous work around processing the data," he adds. "By containerizing each of those workloads, we can scale them up and down independently. Google Kubernetes Engine also allows different teams to work in isolation, which has been beneficial from a development standpoint as well as an infrastructure standpoint."

Maxwell Plus also uses Cloud VPN (<https://cloud.google.com/hybrid-connectivity/>) for a highly secure connection between its systems and data and those of its clinical collaborators, as well as Cloud DNS (<https://cloud.google.com/dns/>) to manage domain names and Cloud Load Balancing (<https://cloud.google.com/load-balancing/>) to support scalability and optimize the use of Google Cloud Platform resources.

Existing literature validated

Patient data processed and analyzed to date has, Dr. Smith says, enabled Maxwell Plus to help validate existing literature that indicates clinicians lack confidence in existing early-stage testing procedures for prostate cancer and other conditions.

This lack of confidence has prompted clinicians to move quickly to the biopsy stage to confirm the validity of their diagnoses. "New technologies have a lot of potential to rectify this situation and guide treatment to be more accurate, specific, and cost-effective," he says.

Results delivered in one to five minutes

Running in Google Cloud Platform has enabled Maxwell Plus to reliably deliver results to clinicians within minutes. "Clinicians are used to getting results back in two days to a week," says Dr. Smith. "With our platform running in Google Cloud, we deliver results in one to five minutes, regardless of the number of patients coming in."

The organization is also reducing false positive cancer diagnoses by 30 percent and allowing men to work with doctors to manage their risk.

Google Cloud Platform has also given the business the ability to accelerate its development cycles, test new ideas easily on a subset of data, test in parallel, and deliver new services considerably faster than in other environments. In addition, the flexible Google Cloud Platform charging model, aligned with the ability to scale compute capabilities quickly and easily, has enabled the fledgling business to control its costs.

"Our expertise
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*—Dr. Elliot Smith, Founder,
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Support growth

With its platform running smoothly in Google Cloud Platform, Maxwell Plus is expanding internationally and now works with clinicians in New Zealand, Europe, and the United States as well as Australia.

"Google Cloud Platform is the only way we can grow

from an Australian-based company to a global company," says Dr. Smith. "If we had the burden of looking to set up our own infrastructure, it simply wouldn't be feasible."

Furthermore, using Google Cloud Platform has enabled Maxwell Plus to meet stringent requirements governing the storage and processing of medical data in various jurisdictions. "By working with Google Cloud Platform, we can access in-country compute and storage and isolate projects very quickly," adds Dr. Smith.

The establishment of the Google Cloud Platform Region in Sydney enabled Maxwell Plus to move compute and storage from the United States or Singapore to the local market, improving performance for clinicians in its home market.

Maxwell Plus' use of Google technologies extends beyond Google Cloud Platform to Google Workspace (<https://workspace.google.com/>). These applications power the company's internal productivity and collaboration activities and according to Dr. Smith, have allowed team members to work at a rapid pace.

Google Cloud Platform is poised to play an integral role in the business's future. "With Google Cloud Platform available, it doesn't make sense for us to use our own infrastructure," Dr. Smith says. "Our expertise complements Google Cloud specialties of machine learning, infrastructure, managed services, and ease of use. We see a bright future ahead in helping to transform healthcare globally."

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